PROBLEM 10 – GPT-6 ASTRA

Abstract. I cannot provide a complete proof of the assertion in the question. The argument below establishes its lattice-theoretic prerequisites and identifies the remaining geometric existence problem. It is not a proof of the requested theorem.

1. Scope and reference conventions

The proposed conclusion specializes the forward existence assertion in [GT24, Conjecture 5.10], with the strengthened positivity hypothesis in the question. The fact that the general assertion is formulated as a conjecture does not, by itself, establish that this specialization is open. Nevertheless, citing that conjecture would not constitute a proof.

Here, references to numbered statements in [GT24] concern arXiv version 3, dated July 22, 2024. Its Conjecture 5.10 is Conjecture 5.12 in version 1.

Write

$$\Lambda = H^{\oplus 3} \oplus E_8^{\oplus 2}, \qquad L = H, \qquad M = H \oplus E_8^{\oplus 2}.$$

For a marked smooth fibre of the degeneration, let $I \subset \Lambda$ be its primitive rank-two isotropic monodromy sublattice, as in [GT24, Section 3.1].

Monodromy invariance gives $I \subset L^{\perp}$. For completeness, over $\mathbb{Q}$, if N is the logarithm of monodromy, then N is skew-adjoint for the intersection pairing and annihilates L. Consequently,

$$(l, Nx) = -(Nl, x) = 0 \qquad (l \in L,\ x \in \Lambda_{\mathbb{Q}}).$$

The same orthogonality holds for the saturation I of the image of N.

2. The automatic lattice-theoretic conditions

Lemma 2.1. *Let T be an even unimodular lattice, and let $I \subset T$ be a primitive, totally isotropic sublattice of rank two. There is an orthogonal decomposition*

$$T \cong H_1 \oplus H_2 \oplus K$$

in which I is generated by one primitive isotropic vector from each of H_1 and H_2. In particular, I is doubly admissible in the terminology of [GT24, Definition 3.31].

Proof. Choose a basis e_1, e_2 of I. Since I is primitive, T/I is free. Thus restriction gives a surjection

$$\mathrm{Hom}(T, \mathbb{Z}) \longrightarrow \mathrm{Hom}(I, \mathbb{Z}).$$

Unimodularity identifies T with $\mathrm{Hom}(T, \mathbb{Z})$. We can therefore choose $w_1, w_2 \in T$ satisfying

$$(e_i, w_j) = \delta_{ij}.$$

Since T is even, the vectors

$$g_i = w_i - \frac{w_i^2}{2} e_i$$

are integral and isotropic, and still satisfy $(e_i, g_j) = \delta_{ij}$.

Put $c = (g_1, g_2)$, and replace g_2 by $g_2 - ce_1$. Its square remains zero because $(g_2, e_1) = 0$. After this replacement, $(g_1, g_2) = 0$, while all the required pairings with e_1, e_2 remain unchanged. Hence

$$U = \mathbb{Z}e_1 + \mathbb{Z}g_1 + \mathbb{Z}e_2 + \mathbb{Z}g_2$$

is isometric to $H \oplus H$.

A unimodular sublattice splits off orthogonally: for each $x \in T$, unimodularity of U supplies $u \in U$ with $(u, v) = (x, v)$ for every $v \in U$, and then $x - u \in U^\perp$. Thus $T = U \oplus U^\perp$, as required. □

Proposition 2.2. *For the degeneration in the question, there is a negative-definite even unimodular lattice K of rank sixteen such that*

$$L^\perp \cong H^{\oplus 2} \oplus K, \qquad I_\Lambda^\perp / I \cong L \oplus K.$$

Under the second isomorphism, the polarization is the indicated summand L. In particular, its embedding in $I_\Lambda^\perp / I$ is primitive, and the degeneration is doubly admissible.

Proof. Because $L = H$ is unimodular, its embedding in Λ splits orthogonally:

$$\Lambda = L \oplus T, \qquad T = L^\perp.$$

The lattice T is even unimodular of signature $(2, 18)$. Apply Lemma 2.1 to $I \subset T$. The remaining summand K has signature $(0, 16)$. In the resulting decomposition, the orthogonal complement of I inside $H_1 \oplus H_2$ is exactly I. Taking the quotient by I therefore gives

$$I_\Lambda^\perp / I = L \oplus K.$$

All the assertions follow. □

Lemma 2.3. *Every primitive vector $e \in I$ is 1-admissible in $L^\perp$. Its mirror lattice is*

$$e_{L^\perp}^\perp / \mathbb{Z}e \cong M.$$

Proof. Set $T = L^\perp$. Since e is primitive and T is unimodular, there is a vector $w \in T$ with $(e, w) = 1$. The vector

$$g = w - \frac{w^2}{2} e$$

is integral and isotropic, with $(e, g) = 1$. Thus $\mathbb{Z}e + \mathbb{Z}g \cong H$ splits off from T. This proves 1-admissibility and shows that $e_T^\perp / \mathbb{Z}e$ is even unimodular of signature $(1, 17)$. The classification of indefinite even unimodular lattices identifies it with $H \oplus E_8^{\oplus 2}$; see, for example, [Nik80, Section 1.14]. □

3. An elliptic fibration with the required abstract frame lattice

The preceding lattice calculation also produces an elliptic fibration on Y whose abstract frame lattice is the correct one. It does not produce a splitting loop.

Proposition 3.1. *Let Y be as in the question, and let K be the lattice from Proposition 2.2. There is an elliptic fibration*

$$f : Y \longrightarrow \mathbb{P}^1$$

with a section and fibre class F, such that

$$F_{\mathrm{NS}(Y)}^\perp / \mathbb{Z}F \cong K.$$

Proof. Very generality gives $\mathrm{NS}(Y) = M$. The lattice $H \oplus K$ is even unimodular of signature $(1, 17)$; hence it is isometric to M.

Choose a primitive isotropic generator of the factor H in this decomposition of $\mathrm{NS}(Y)$. After changing its sign and applying reflections in classes of (-2)-curves, it becomes a primitive nef class F. The standard reduction to the nef cone and the elliptic-pencil theorem for primitive nef isotropic classes give an elliptic fibration with fibre class F; see [Huy16, Chapter 8, Section 2, and Chapter 11, Section 1]. Since these reflections are lattice isometries,

$$F_{\mathrm{NS}(Y)}^\perp / \mathbb{Z}F \cong K.$$

We check the existence of a section. Unimodularity supplies $w \in \mathrm{NS}(Y)$ with $(F, w) = 1$. Replacing w by $w - \frac{w^2}{2}F$, we may suppose that $w^2 = 0$. Then

$$D = w - F \quad \text{satisfies} \quad D^2 = -2, \qquad D \cdot F = 1.$$

Riemann–Roch on a K3 surface implies that D or $-D$ is effective. The latter is impossible because F is nef and $(-D) \cdot F = -1$. An effective representative of D therefore has a horizontal irreducible component of degree one over $\mathbb{P}^1$. Its normalization gives a section. □

This is the coarse degeneration–fibration correspondence: it matches the relevant abstract lattices, but not the componentwise splitting required in the question. Compare [DHT17, Proposition 4.1].

4. The missing geometric assertion

Suppose that an allowable loop γ for the fibration in Proposition 3.1 has been constructed and that its associated class τ lies in $\mathrm{NS}(Y)^\perp$. Put

$$Q_X = I_\Lambda^\perp / I, \qquad Q_Y = (\mathbb{Z}F \oplus \mathbb{Z}\tau)^\perp / (\mathbb{Z}F \oplus \mathbb{Z}\tau), \qquad \Gamma = F_{\mathrm{NS}(Y)}^\perp / \mathbb{Z}F.$$

Use the component sublattices from [GT24, Definition 5.1]:

$$P_i^X = \varphi_X(K_{V_i}^\perp), \qquad P_i^Y = \varphi_Y([r_i(b)]^\perp).$$

Here r_i is the right adjoint of the asymptotic charge map of the fibration over the i-th disc.

The two polarizations are then doubly admissible. Indeed, this has already been proved for the degeneration. On the fibration side, the section makes F 1-admissible. Moreover, $\mathrm{NS}(Y)^\perp \cong H^{\oplus 2}$ is unimodular, so the primitive isotropic class τ is 1-admissible by the argument of Lemma 2.3.

Consequently, [GT24, Proposition 5.3] reduces the mirror-pair condition to the existence of an isometry

$$(*) \qquad \psi : Q_Y \xrightarrow{\sim} Q_X \quad \text{such that} \quad \psi(\Gamma) = L_{Q_X}^\perp, \qquad \psi(P_i^Y) = P_i^X \quad (i = 1, 2).$$

The unresolved requirements are therefore:

(1) construct an allowable loop compatible with the polarization;
(2) verify the componentwise equalities in $(*)$ for the *given* Tyurin model.

Allowability includes both a specified boundary-monodromy condition and a primitivity condition; see [GT24, Assumption 4.9 and Definition 4.11].

Neither requirement follows merely from the abstract isomorphism $\Gamma \cong K$. In particular, an isometry of the ambient lattices need not identify the two distinguished component sublattices.

The Type II constructions in [ABE22, Section 7F] provide standard divisor models. Using them here would require an additional comparison with the given polarized Tyurin model, followed by the loop construction and the integral componentwise verification above.

Conclusion

The calculations above verify the mirror lattice, double admissibility, and the existence of an elliptic fibration with the correct abstract frame lattice. The sources also provide the precise conditional criterion for being a mirror pair.

I have not established the required allowable-loop construction and componentwise matching for every degeneration satisfying the strengthened hypotheses. Thus a complete proof has not been achieved, and presenting the preceding reduction as one would leave the central geometric assertion unproved.

References

[ABE22] V. Alexeev, A. Brunyate, and P. Engel, *Compactifications of moduli of elliptic K3 surfaces: Stable pair and toroidal*, Geom. Topol. **26** (2022), no. 8, 3525–3588.

[DHT17] C. F. Doran, A. Harder, and A. Thompson, *Mirror symmetry, Tyurin degenerations and fibrations on Calabi–Yau manifolds*, String-Math 2015, Proc. Sympos. Pure Math., vol. 96, American Mathematical Society, 2017, pp. 93–131.

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